Rotaract Club of Xiamen University Malaysia

Rotary Club of Kuala Lumpur Dynamics

Rotary Club of Tun Razak Exchange

SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named “SDC2026_TeamName_Preliminary.pdf” before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

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1. TEAM AND PROJECT OVERVIEW

Team Name

DOVUN

Selected Domain

Domain One: Aiversecity Future-Ready Workforce

Selected SDG Track (1–17)

SDG 8 (8.5); supporting SDG 4 (4.4, 4.3)

Project / Solution Name

AIVANCIN

Name (Leader)

Jad Fahmi

Email & Phone (Leader)

jadfahmi07@gmail.com (phone in the sdchack.my submission)

University Affiliation(s)

German-Malaysian Institute

Number of Team Members

5

2. EXECUTIVE SUMMARY & PROBLEM STATEMENT

2.1. The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

In Q2 2026, 40.7% of employed tertiary-educated Malaysians aged 25-34 worked in semi-skilled or low-skilled occupations: 751,100 people, according to DOSM [1]. AIVANCIN focuses on these graduates when they are deciding whether a new career path is worth their evenings and savings.

A job title can hide relevant work. A delivery driver may also troubleshoot laptops, manage a family business website or keep financial records. Selecting training from a job title alone can overlook that experience, while a list of attractive occupations does not explain the missing qualifications, practical tasks or cost of starting.

Changing AI use adds another question: which activities should a learner practise doing, and which should they learn to direct and check? Global AI-usage evidence and US job projections can inform exploration, but neither establishes Malaysian hiring demand or an individual's employment prospects.

The immediate need is a small, explainable set of roles to explore and an affordable first learning block, with sources and entry barriers visible. The project supports skills development and decent work under SDG targets 4.4 and 8.5 [2]; it does not promise placement or a reduction in national underemployment.

2. EXECUTIVE SUMMARY & PROBLEM STATEMENT

2.2. The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

AIVANCIN is a web application that connects work described in plain English to graduate-level career routes. A local all-MiniLM-L6-v2 sentence encoder, fine-tuned by the team on a RunPod GPU, ranks O*NET work activities by meaning [3,6]. It highlights the source sentence, and the user confirms only activities that are true. Nothing is accepted silently; a guard holds back negated, planned or other people's work.

Confirmed activities feed reproducible overlap matching. Users compare up to three routes among 17 supported roles, inspect missing activities and separate qualification or experience requirements, and choose a destination. An ordered first four-week learning block uses verified free resources in prerequisite order, within their available time and budget. Uncovered gaps remain visible.

The Future Skills Radar keeps four signals separate: published US job outlook, demand carried to activities through task links, global Claude usage, and observed Malaysian employment/pay [1,3-5]. Slower, moderate and faster scenarios show estimated AI-use shifts. A checkpoint adds AI checking only when an activity crosses a visible threshold. Every plan step carries reasoned SDG tags. What changed compares a plan under two evidence releases, and the user keeps it or accepts the update. A cohort view shows career centres the shared gaps of a synthetic cohort.

All of this runs in the live prototype on Replit. The team did not attend the Phase 1 workshops. We therefore claim no workshop-acquired skills; implementation uses independently applied API design, structured data and cloud-deployment techniques.

3. TECHNICAL ARCHITECTURE & IMPLEMENTATION PLAN

3.1. Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

CURRENT STACK (LIVE ON REPLIT)
Browser: HTML, CSS and JavaScript.
Backend: Python 3.13, FastAPI and Uvicorn.
Inference: all-MiniLM-L6-v2, fine-tuned once on RunPod, quantized ONNX, CPU.
Storage: bundled CSV, JSON and NumPy snapshots; no database service. No external model API is required.

SYSTEM FLOW
Browser description -> FastAPI -> sentence encoder
-> activity suggestions -> human confirmation
-> overlap matcher + dated evidence -> routes and gaps
-> resource catalogue + constraints -> learning block.

PUBLIC DATA
O*NET 31.0, Anthropic Economic Index, DOSM, BLS outlook via O*NET OnLine, NCES and ESCO crosswalks [1,3-7]. A curated catalogue links Microsoft Learn and Saylor University resources [8].

SPONSOR USE
Replit: the live prototype, aivancin--jadfahmi.replit.app. RunPod: one RTX 4090 pod fine-tuned the encoder on O*NET task/activity pairs (US$0.42); shipped because it passed a gate fixed before training (45 vs 39 of 60 held-out). No sponsor API calls.

3. TECHNICAL ARCHITECTURE & IMPLEMENTATION PLAN

3.2. Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

We framed one decision for a specific user: an underemployed graduate choosing a career direction before paying for retraining. Public DOSM evidence established the problem; synthetic personas let us examine the journey without collecting real CVs or personal work histories.

The prototype exposes suggestions for confirmation, supports role exploration before intake, and shows one review sentence at a time. Route summaries reveal requirements before a learning commitment. Rendered checks at phone and desktop viewport sizes cover light/dark appearance, navigation and saved progress. No interviews or real-user usability validation have been completed.

Offline comparisons showed a useful encoder contribution but weak matching in generic fields. The hardest risk is relevant routes with genuinely resourced learning blocks. We expanded finance/data coverage to 17 supported roles, keep explicit incomplete states and test each change on a fresh predeclared profile set.

The app runs on Replit from bundled data. Missing inference assets trigger labelled keyword fallback; changed data hashes stop startup. The same model and snapshots run offline as a demo fallback. Scenarios and evidence updates preserve completed work; claims stay bounded to executed checks.

4. PRELIMINARY ENVIRONMENTAL & SDG IMPACT FRAMEWORK

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess Innovation and Creativity as well as Feasibility of build. Vague or unsupported projections may weaken your case for advancing to Phase 2.

4. PRELIMINARY ENVIRONMENTAL & SDG IMPACT FRAMEWORK

Metric Category	Your Response
Target SDG Sub-Target(s) *Identify the exact UN SDG sub-target your project addresses* *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	4.4: relevant skills for employment and decent jobs. Activities missing from a chosen route become explicit learning goals. 8.5: productive employment and decent work. Show skilled-role destinations, pay context and entry barriers. 4.3: affordable technical learning, supported by free resource links. These are contribution mechanisms, not measured employment outcomes [2].
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?* *e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	National problem: 40.7%, or 751,100 employed tertiary-educated Malaysians aged 25-34, were in semi-/low-skilled occupations in Q2 2026 [1]. Product evidence: across five predeclared synthetic sets, the current version qualifies 35 of 52 in-scope profiles (67.3%). The two newest sets, on their first run, gave 8 of 12 and 1 of 4 [9]. No real-user employment or learning-completion baseline has been measured.

4. PRELIMINARY ENVIRONMENTAL & SDG IMPACT FRAMEWORK

Metric Category	Your Response
Core Measurable KPI *The specific unit you will track* *e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided*	Route-and-block coverage (%): 100 x qualifying in-scope synthetic profiles / all predeclared in-scope profiles. A profile qualifies only with at least two relevant skilled-role routes. Each needs sourced entry requirements and a four-week block serving at least three priority gaps with verified resources within time and budget. Report out-of-scope cases separately. This is a prototype-quality proxy, not an SDG outcome [9].
Phase 2 Impact Target *Your realistic, percentage-based target by the final prototype presentation* *e.g., "20% improvement in computational resource use."*	At least 80% route-and-block coverage on a fresh, predeclared in-scope synthetic profile set by the final prototype presentation [9]. Freeze profiles, relevance criteria and constraints before changes; measure the current version on that identical set, then repeat. Keep every failure in the denominator. This is the adopted engineering target, not a predicted reduction in national underemployment. Real employment and course-completion outcomes remain unmeasured.

5. ELIGIBILITY & READINESS CHECKLIST

Tick all boxes that apply before submitting.

☐ All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.
Not completed (confirmed by the team leader, 21 October 2026)

☑ The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.

☑ The proposed solution targets a functional, deployed application rather than a theoretical business case study.

☐ All fields in this template have been completed within the word limits stated.

Narrative responses are within the word limits. Workshop completion is not claimed: the team did not attend. The last box is ticked by the team leader when signing.

6. IP & ORIGINALITY DECLARATION

6.1. Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☑ Yes ☐ No

If yes, briefly describe what was pre-existing and which team member(s) owned it:

Yes. Known pre-existing material:

1. KL Pro typeface, created by Jad Fahmi for DOVUN before the event (October 2026). Its unmodified files supply typography. It retains a separate, all-rights-reserved licence and is not covered by the product's Apache 2.0 licence. Distribution rights for the product were confirmed by Jad.

2. The team's README/document structure prepared before the event, adapted alongside the official SDC sample README. This is documentation structure, not pre-existing application functionality.

Third-party models, libraries, datasets, crosswalks and fonts are listed in section 6.2 and Appendix B. The submitted product code is maintained in the event repository.

6. IP & ORIGINALITY DECLARATION

6.2. Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

☑ Yes ☐ No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	Type (code/dataset/model)	License
O*NET + DOSM	Workforce datasets; job outlook	CC BY 4.0
Anthropic Economic Index	Global AI-usage datasets	CC BY
all-MiniLM-L6-v2	Pre-trained sentence encoder, fine-tuned by the team	Apache 2.0
Runtime and test libraries	FastAPI, Uvicorn, ONNX Runtime, Tokenizers, NumPy, pytest, HTTPX	MIT; BSD-3-Clause; Apache 2.0; bundled notices
Crosswalks + Jost	NCES, ESCO; font. Detail: Appendix B.	Public domain; EC reuse; OFL 1.1

6. IP & ORIGINALITY DECLARATION

6.3. Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes ☑ No

If yes, please describe the nature of the agreement and which member it concerns:

None. Jad Fahmi confirmed on 21 October 2026 that no team member has an employment, internship or consulting agreement under which an employer could claim the hackathon work.

7. DECLARATION

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.

[Signature]

[Date]

Appendix A: Sources and evidence

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[1] Department of Statistics Malaysia. **Quarterly Skills-Related Underemployment by Age**, Q2 2026. Rate 40.7%; 751.1 thousand people for ages 25-34. CC BY 4.0. open.dosm.gov.my/data-catalogue/lfs_qtr_sru_age

[2] United Nations, Department of Economic and Social Affairs. SDG targets 4.3 and 4.4: sdgs.un.org/goals/goal4. Target 8.5: sdgs.un.org/goals/goal8. Descriptions in section 4 are paraphrases.

[3] USDOL/ETA. **O*NET 31.0 Database**, files dated 25 August 2026; CC BY 4.0. onetcenter.org/database.html. AIVANCIN uses modified extracts. USDOL/ETA has not approved, endorsed or tested these modifications. O*NET is a trademark of USDOL/ETA.

[4] Anthropic. **Economic Index**, bundled releases 15 September 2025, 15 January 2026, 24 March 2026 and 26 June 2026. CC BY release notices. anthropic.com/economic-index. Usage describes global Claude conversations, not Malaysian labour demand.

[5] O*NET OnLine, BLS US 2025-2035 job projections: onetonline.org/find/bright. Malaysian observed employment/pay: open.dosm.gov.my/data-catalogue/wages_occupation. US projections and Malaysian group observations remain separate.

[6] sentence-transformers. **all-MiniLM-L6-v2**, pinned revision 1110a243, quantized ONNX. Apache 2.0. huggingface.co/sentence-transformers/all-MiniLM-L6-v2.

[7] NCES, CIP 2020 to SOC 2018 crosswalk (US federal work): nces.ed.gov/ipeds/cipcode. European Commission, ESCO O*NET-ISCO crosswalk; reuse with attribution under Decision 2011/833/EU: esco.ec.europa.eu.

[8] Linked learning providers: Microsoft Learn and Saylor University. Links and provider objectives are curated; missing coverage is explicit. Resource availability is checked when catalogued, not guaranteed indefinitely.

[9] Team offline evaluation, recorded 21 and 22 October 2026 (python tools/checks/b7_target.py). Route-and-block coverage: sets 1 to 5 give 9, 10, 7 and 8 of 12 and 1 of 4 in-scope profiles (35 of 52); sets 1 and 2 guided earlier changes. Adopted Phase 2 target: at least 80% on a fresh predeclared in-scope set, with both builds evaluated on identical inputs. Synthetic relevance and route coverage are engineering proxies. No real-user benefit or employment effect has been established. Repository: github.com/AIVANCIN/aivancin. Live prototype: aivancin--jadfahmi.replit.app.

Appendix B: Resource and licence detail

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Runtime code: FastAPI and ONNX Runtime, MIT; Uvicorn and NumPy, BSD-3-Clause; Tokenizers, Apache 2.0. NumPy also contains bundled components under 0BSD, MIT, Zlib and CC0 notices. Those notices remain applicable.

Test code: pytest, MIT; HTTPX, BSD-3-Clause. These support offline verification and are not external product APIs.

Model: all-MiniLM-L6-v2 from sentence-transformers, Apache 2.0, fine-tuned once by the team on a RunPod RTX 4090 with O*NET 31.0 task-to-activity pairs, then quantized to ONNX. It shipped because it passed a gate fixed before training: 45 of 60 held-out descriptions against 39 for the original model.

Datasets: O*NET 31.0, O*NET OnLine outlook extracts and DOSM labour/pay datasets, CC BY 4.0; Anthropic Economic Index, CC BY as recorded in each release. Source releases, access dates, modifications and hashes are recorded in the repository data manifest.

Crosswalks: NCES CIP-SOC, US federal government work (public domain); European Commission ESCO O*NET-ISCO mapping, attribution and reuse under Commission Decision 2011/833/EU.

Fonts: Jost by Owen Earl and the Jost Project Authors, SIL Open Font License 1.1. KL Pro by Jad Fahmi for DOVUN, pre-existing, all rights reserved, separate licence. KL Pro is not included in the project Apache 2.0 grant.

Documentation structure: official SDC sample README structure, adapted alongside the pre-event team README structure. The official sample repository is distributed under Apache 2.0. No sample application functionality is claimed as original AIVANCIN functionality.

Course resources: external links and factual metadata, rather than copied course content. Microsoft material remains subject to its terms. Saylor-authored content is CC BY 3.0, excluding final exams and separately licensed third-party content. AIVANCIN does not redistribute course lessons.

Current limitations: English intake; US occupation and requirement data; Malaysian pay/employment at broad group level; weak route coverage in generic fields. Moderate AI-use outlook backtest MAE: 16.33 percentage points. Slower/faster scenarios are assumptions. They are not Malaysian employment forecasts.

Team status: the leader supplied contact details and confirmed no workshop attendance and no employment-related IP agreements on 21 October 2026. Workshop completion is left unticked. Sponsor use (Replit deployment, RunPod fine-tuning) is declared, with proof, in the Technical and Tools Declaration.